

Low Yu Xuan

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Personal Profile

Computer Science student specializing in Data Science and Artificial Intelligence, with hands-on experience in backend development and applied machine learning across computer vision, NLP, and predictive modeling. Award-winning competitor in international AI Olympiads with a track record of building end-to-end ML systems alongside production backend services. Combines strong technical depth with proven leadership across cross-functional and international teams. Seeking software development and AI/ML research opportunities to build intelligent, real-world systems.

Education

University College Dublin Sep 2026 – Present
Computer Science with Data Science and Artificial Intelligence

INTI International College Penang Apr 2025 – Jun 2026
Cambridge A Levels — 4A (Mathematics, Further Mathematics, Physics, Chemistry)*

Chung Ling High School Penang Jan 2020 – Dec 2024
Sijil Pelajaran Malaysia (9A+, 1A) — Ranked 2/441 in Year, Best Graduate of 2024

Leadership Experience

Malaysian AI Olympiad Fellowship, [MAIO](#) Aug 2026 – Present

- Plan 14+ trainings for Malaysian students on Olympiad AI, involving ML, CV, NLP and Audio domains
- Brainstorm and refine 2+ task ideas to ensure Malaysian students are trained and exposed to challenging problems

Vice President of Activities, 21st INTIMA, INTI Nov 2025 – Present

- Review and manage 84+ events across campus by societies and 28+ events and initiatives by INTIMA
- Ensure wellbeing and constitutional compliance of 37 societies; established 2 new societies in students' interests

Technical Director, [AMISO](#) Jun 2025 – Present

- Coordinated a team of 5 developers across Malaysia to expand outreach and maintain 100% website uptime
- Developed a blueprint for a mobile application to promote International Science Olympiads

Section Leader, [Code in Place](#) 2025 Apr 2025 – Jun 2025

- Taught 15+ students Python from all around the world from scratch
- Conducted 2 TeachNow sessions to help students identify bugs and improve programming skills

Skills

- Programming Languages:** Python, C++, JavaScript, SQL, PHP
- Machine Learning & AI:** Computer Vision, NLP, Deep Learning, Model Training & Fine-Tuning, Data Pipeline Engineering, Model Evaluation & Benchmarking
- Software Development:** Backend Development (Django), REST API Design, MVC Architecture, Mobile Development (Flutter)
- Professional:** Technical Leadership, Cross-Functional Team Coordination, Research Communication, Public Speaking, Problem Solving

Awards and Honors

- International Olympiad in Artificial Intelligence – **Bronze** (2026)
- Asia Pacific Olympiad in Artificial Intelligence – **Honorable Mention** (2026)
- National Artificial Intelligence Competition (NAIC) – **1st Runner Up** (2025), **Champion** (2026)
- Northern Eurasia Olympiad in Artificial Intelligence – **Rank 4/188 (Day 1)**
- Malaysia Artificial Intelligence Olympiad – **Gold** (2025), **Rank 1** (2026 Qualifiers)
- USA Computing Olympiad – **Gold Division** (2026)

Certification

- Neural Networks and Deep Learning, DeepLearning.ai May 2025
- Project Management Fundamentals Jun 2025

Projects

GUIDE-SHARK

- Frontend application written in Flutter using clean code architecture for readability
- Backend application written using Django and provide Rest API services, applying SoC design principle
- Applied Artificial Intelligence (AI) to analyze symptoms and provide quick diagnoses

JotMe

- Frontend features multiple dynamic mood visualization through ChartJS and interactive dashboard
- Backend system integrating KeyBERT for keyword extraction and pre-trained emotion detection models, implementing graph-based contextual analysis for sentiment scoring
- Applied NLP to analyze journal entries to provide personalized mental health insights with crisis intervention features

Byte of Kuih

- Collected 10000+ images to train EfficientNetV2 model on recognising Malaysian traditional kuih
- Augmented and cleaned data to construct a consistent dataset containing high-quality samples (16000+)
- Fine-tuned model to achieve 98% validation accuracy and 100% testing accuracy

Embrace Study

- Prospective cohort study examining the impact of maternal-infant postpartum psychological adjustment
- Developed the trackers and ensured derived insights are meaningful and interpretable with robust trackings and secure management to ensure no data leaks
- Developed with Next.js and ShadCN with Supabase as the backend, targeting 200+ participants

Diabetic Retinopathy Screening

- Optimized preprocessing pipeline that uses CLAHE, Graham's Method and Feature Adaptive Cropping
- Trained an ensemble model (ResNet-34, EfficientNetB4, RandomForest) that reaches 91% accuracy on unseen data
- Optimized training with LRSchedulers and Gradient Norm Clipping while working with limited unbalanced data